

June 12, 2024

1 ALGORITHM

Step 1: Input

Ask the user to select the number system they want to convert from (Decimal, Binary, or Octal) Store the user's choice in choice1 Ask the user to enter a number in the selected number system Store the entered number in num Ask the user to select the number system they want to convert to (Decimal, Binary, or Octal) Store the user's choice in choice2

Step 2: Conversion

Use a nested switch statement to perform the conversion based on choice1 and choice2

Case 1: Decimal to ...

If choice1 is 1 (Decimal), perform the following conversions: If choice2 is 1 (Decimal), print a message indicating that the number is already in decimal If choice2 is 2 (Binary), convert the decimal number to binary using a loop and print the result If choice2 is 3 (Octal), convert the decimal number to octal using the oct manipulator and print the result

Case 2: Binary to ...

If choice1 is 2 (Binary), perform the following conversions: Ask the user to enter a binary number Convert the binary number to decimal using a loop If choice2 is 1 (Decimal), print the decimal representation of the number If choice2 is 2 (Binary), print a message indicating that the number is already in binary If choice2 is 3 (Octal), convert the decimal number to octal using the oct manipulator and print the result

Case 3: Octal to ...

If choice1 is 3 (Octal), perform the following conversions: Ask the user to enter an octal number Convert the octal number to decimal using a loop If choice2 is 1 (Decimal), print the decimal representation of the number If choice2 is 2 (Binary), convert the decimal number to binary using a loop and print the result If choice2 is 3 (Octal), print a message indicating that the number is already in octal

Step 3: Output

Print the result of the conversion to the console

Step 4: Terminate

Return 0 to indicate successful execution of the program